What evidence is there of Gender Bias in Textbook Illustrations in India

Candidate Number : 43958

Supervisor : Tom Robinson

Date : August 28th 2025

Word Count : 9500 words

TABLE OF CONTENTS

ABSTRACT 3

ACKNOWLEDGEMENTS 4

CHAPTER 1: INTRODUCTION 5

1.1 Introduction 5
1.2 Literature Review 5
1.3 Research Questions 7
1.4 Research Design 8

CHAPTER 2: METHODOLOGY 9

2.1 Manual Annotation Procedure 9
2.2 Image Embeddings 10
2.3 Computational Methods 10
2.3.1 Gender Classification 10
2.3.2 Scene/Place Classification 12
2.3.3 Action Recognition 13
2.3.4 Clothing Classification 15
2.3.5 Color Analysis 18

CHAPTER 3: RESULTS 18

3.1 Exploratory Results 18
3.2 Statistical Tests 24

CHAPTER 4: CONCLUSION 28

REFERENCES 29

ABSTRACT

This dissertation investigates gender bias in Indian primary school textbooks (grades 1–5) by analysing 3,300 illustrations through an integrative computational pipeline. Prior work on textbook bias has been largely qualitative and text-focused; our contribution lies in scaling visual analysis with computer vision and zero-shot models. We find women underrepresented at a ratio of nearly 2:1 compared to men, confirming long-observed disparities now echoed in national-scale evidence. Spatial classification show women disproportionately confined to indoor settings such as kitchens and classrooms, while men dominate outdoor and professional settings. Inference on action and pose reveal men more frequently potrayed in active or leading roles, whereas women appear in supportive or passive positions. These results demonstrate how visual depictions in textbooks reproduce structural gender hierarchies, highlighting the urgency of curriculum reform toward gender-just education.

ACKNOWLEDGEMENTS

For the girl child who once asked me “Do you know how time fits into clocks”.

To Tom Robinson and Dan De Kadt, who taught me that statistics could be both rigorous and rebellious; to Olivia from ASDS, who answered my panicked emails with the patience of a saint; to Mum and Dad, who still aren't entirely sure what computational social science is but are terrifically supportive nonetheless; and to my roommate Shubha, who always kept the coffee coming, this dissertation owes its existence to your collective tolerance for my academic dramatics and your unwavering faith that somewhere beneath the scatter plots lay something worth saying.

INTRODUCTION

Gender is one of "a microcosm of ideologies, values, and beliefs" (Taylor, 2003:301).

From cover to cover of the Indian 10th grade economics textbook used nationwide, every image of a businessman…is of a man. Gender bias in textbooks (GBIT) has been a well-documented problem globally and especially in Southern Asia where Confucian views on gender roles still prevail.
This research design aims to answer the question: "What evidence is there of gender bias in the illustrations in Indian textbooks aimed at grades 1-5". We justify this line of questioning by arguing -in order- that the static lattices of gender norms, roles and attitudes are socially learnt instead of intrinsically assigned, that early literature consumed contributes to development of these concrete associations of gender and roles, and that a picture superiority effect means illustrations are stronger than words at reinforcing these associations.
These matrices of gender associations then percolate into implicit attitudes to form entrenched structural biases that inform how gender interacts with agency and power to create systems of oppression.
We argue simply that representation and depiction matter, and that bias nested in the strongholds of education dissemination – textbooks - are counterproductive to building equitable societies. We hence seek to fill a knowledge gap in the literature by using an integrative multi-method computational social science approach to look at what forms of gender bias exists in Indian textbook illustrations. We pursue structured methods of enquiry employing appropriate deep learning models to study places, position, action, clothing are assigned across gender. We also employ BLIP2, prompt engineering and ViBE to pursue exploratory lines of questioning and use unsupervised learning methods to surface interesting patterns. To underscore the relevance of our potential findings, we finish by highlighting both the individual implications of gender bias and position the relevance of our findings in the feminist political economy and critical pedagogy scholarship.

LITERATURE REVIEW

The 2005 Millennium Development Goals Report flagged that the content of textbooks by casting in women in subordinate roles could reinforce existing gender stereotypes. (UNDP, 2005). The SDG 5 hence supports the production of curriculum free from gender stereotypes.

In the Indian context, the following evidence of GBIT has been documented across text and illustrations. Dipta Bhog (2002) records men being depicted with greater spatial mobility – a primary limiter of women's agency in India- and women depicted as naïve. Of the 75 lessons she reviewed only 8 positioned women as achievers while over 50% featured only male actors in text. Kalia (1980) also finds a 2:1 ratio of male to female centric stories in the Hindi NCERT Textbook. In Hyderabad, of 13 English textbooks the same ratio is 81:9. A 2014 report by NCERT reveals in it's analysis of 18 primary textbooks that English textbooks features primarily boys and while men were depicted in a range of roles, women were overwhelmingly depicted as homemakers, teachers and nurses. (NCERT, 2017). There were also subtler imbalances in group dynamics - in a poem

‘patang’ the boys flew kites while the girls watched.(ibid). In the state curriculum too, in West Bengal woman were painted as passive and powerless and in Tamil Nadu boys were depicted competing in outdoor games with high levels of competition: football, cricket etc while girls were restricted to indoor activities like drawing, hopscotch etc. (Amruthraj, 2012).

STUDY TITLE	METHOD	SHORTCOMING
Dipta Bhog (2002)	Qualitative content analysis	Only deals with grade III,V,VIII
Nandi et al (2020)	Mixed methods qualitative analysis	Grades V textbook
NCERT (2017)	Qualitative and quantitative techniques – content analysis	Standard I-V
Kalia (1980)	Content Analysis	41 Hindi or English language instruction textbooks

The table summarises that while there have been initial probes into specific textbooks, there has been no national scale quantitative analysis of textbooks across grades and subjects focusing on illustrations and using quantitative social science. This area largely remains understudied.

Gender is a sociocultural construct and remains a pervasive global social category. To justify our pursuit of this knowledge gap and illustrate how and why entrenched bias in textbook images matter, we briefly explore how gender bias is introduced and internalised in children.

Developmental Intergroup theory develops the idea that children are intrinsically predisposed to decode the world around them and actively seek cues to evolve categories to classify others (Bigler & Liben, 2007) that can in turn lead to early formation of stereotypes that persist well into adulthood. (ibid) . Cues are fed to them socially and children are exceptionally good at picking up on them and using complex networks of associations to inform behaviour and organise memories.
In the 1970s -the gender schema theories that posit that children develop naïve theories or gender schemas about the attributes of males and females (Bem 1981) and once these networks of gender-linked associations have taken root can inform stereotypes that persist into adulthood. Gender schema theory hence incorporates both social learning theory as well as cognitive theory as schemas serve to organise and interpret new information and develop ‘internalised pattern encoding’ that effect how individuals see the world. The relevance of this recognition is that gender is ‘taught’ and these absorbed patterns or schematic associations are the same that inform implicit biases that balloon and linger into adulthood. The recognition that gender itself and associated bias can be taught, justifies investigation into how we design those teaching materials for young children, that dovetails into our review of the role books, literature early media play.

Bem (1981) finds the gender schema theory dictates that children come to view themselves in terms of ‘gender attributes they acquire through real life and no less through picture books’. Pictures books and illustrations play a central role in children’s cognitive and sociocultural development. Huck (2006) highlights how colourful

illustrations have the ability to shape how children see the world and (Nikoljeva 2010) supports that they can inform their self-perception and help build a complex matrix of intersecting ideas, beliefs and values. Johnson (2019) documents that children are gullible to accepting stereotyped representations. While teaching materials play a part in establishment of association said associations are formed quicker through image data.

The differential upper hand of images in solidifying perceptions of bias is best encapsulated in the picture superior effect (PSE). Baadte et al (2019) find a memory advantage for pictures as compared to words with semantic relatedness of pictures evoking deeper semantic processing of associated items and triggering associative memory. This associative memory is fundamental to the acquisition of knowledge and everyday function and (King et al 2021) highlight that it provides strong impetus for us to develop education materials to account for the outsized effects of picture stimuli on associative recall. Pictures in everyday picture books provide kids with stockpiles of images for their first 'mental museums' that inform the earliest ideas of race, class and gender (Spitz, 1999).

We use these findings from child development literature to argue that while these studies focused on pictures in books, advertisements, children's literature, or experimental design, we argue that textbook illustrations are equally relevant on the same grounds. For starters textbooks are employed, accessed and distributed more widely and in the case of unified national curriculums are used country-wide as opposed to picture books where even the most popular ones may be read by only a small demographic. Furthermore (Sadker and Zittleman, (2007) find "students spend as much as 80 to 95 per cent of classroom time using textbooks and teachers make a majority of their instructional decision based on the textbooks".

In such that we accept that textbook illustrations can introduce and reinforce biases and help form gender schema by parity of reasoning, we can infer they must serve as powerful touchpoints to effect reform (Gleeson et al 2015). Finally, while the majority of associative links and gender schema are evolved between early ages of 5-7, Trautner at el (2005) describes 3 phases of stereotype cementing where after a peak of rigid association between 5-7 years, a phase of relative flexibility follows where ideas can be moulded – this justifies our focus on textbook illustrations beyond the years of 5-7 to the whole of primary school grades 1-5.

RESEARCH QUESTIONS

From the discussion above, our key research question then emerges as : "What evidence is there of gender bias in illustrations in Indian textbooks?". We are interested in enquiring about what forms of gender bias persist. This is an open-ended question that allows us to comment on nuances generated from our analysis. To operationalize this inquiry while maintaining methodological rigor, we articulate four interconnected sub-questions, each calibrated to capture distinct dimensions of gendered representation while collectively constructing a comprehensive portrait of bias.

i. **Do women feature as frequently as men in illustrations in Indian textbooks?** – frequency counts were one of the earliest measures of quantitative content analysis and still provide key insights into representation. Frequency of occurrence by itself though is an incomplete metric. It

leaves out considerations like the settings, role or context in which women are depicted to which end we ask:

ii. **In what settings are men and women depicted in illustrations in Indian textbooks?–** spatial mobility is a pressing concern for women in India closely related to gender roles that dictate women remain at home for childcare and to handle housework. This leads to reduced participation in society, reduced access to information networks, and agency (Mayeux 1995). We capture information on an indoor/outdoor binary, as well as more nuanced hierarchy of spatial occurrence.

iii. **What actions do men and women most commonly perform in illustrations in Indian textbooks?** We aim to capture insights like "are men seen driving more often than women? in group settings are they seem pointing, talking, leading more often? Are women seen more often in care roles or in gendered contexts like fetching water etc? Are they seen playing as often, are there stereotyped patterns in the forms of play?

iv. **What clothing and colours are most by different genders?** Color functions as what Meister et al. (2022) term a "pre-linguistic gender marker," absorbed before children develop critical capacity to interrogate such associations. The pink-blue binary, while seemingly trivial, instantiates deeper logics of differentiation that extend from clothing through toys to entire affective repertoires deemed appropriate for each gender. Yet we must also attend to how clothing signifies beyond mere chromatic choice: the mobility afforded by pants versus skirts.

RESEARCH DESIGN

We adopt an integrative multi-method computational social science approach. Using multiple computational methods when working with image data is a common approach to capture layers of insights (Jang et al 2019).

We narrow our scope by focusing on India and textbooks from grades 1-5 as i. There is some evidence of gender bias though limited in scope and scale, ii. There is one centrally allocated board and syllabus followed and adopted across the country. iii. Our focus on grades 1-5 is justified both by those being crucial years where ideas take root and can be challenged, but also by the fact that senior textbooks often have fewer illustrations. For eg, high school maths and science textbooks have hardly any illustrations (diagrams are not illustrations) and political science textbooks largely feature historical figures and scenes, while they may reflect the choice of stories and information presentation, they are not illustrations. A historical scene may well depict women in stereotyped settings or roles but this may be defensible and necessary on the grounds of historical accuracy. As such we leave these out as they do not reflect active choice of portrayal on behelf of the illustrator or curriculum designer.

This inquiry is motivated by a basic tension, our use of binary gender classification, which classifies figures as "male" or "female" based on visual cues, allows for systematic study while also upholding the gender essentialism that we are criticising.

A primary ethical concern here is that we adopt a binarized perceived gender expression and our categorisation into ‘male’ and ‘female’ grants the notion of binary gender legitimacy and excludes valid representation of non-binary or trans communities. This is an ethical compromise that should be explicitly acknowledged, not just a technological restriction. Our classification reflects perceived gender expression rather than identity, a distinction that is significant but methodologically unachievable given the limitations of computational vision analysis. Gender identity cannot and should not be deduced from visual presentation alone.

There is also evidence of models trained on biased datasets developing biases themselves (Meister et al 2020). They may classify a place as a kitchen because a woman is in it, which is counterproductive and both an ethical and experimental confounder. We spent considerable time addressing this limitation while choosing models. For e.g, to mitigate this, we mask out the bounding box of the person while conducting spatial analysis.

METHODS

Incisive analysis on bias necessitates clean data. Since we rely on machine learning models to generate data we analyse in our results section, we spend considerable time and effort in this section on models, backbones, parameters, ablations and tests use to arrive at the ‘best’ models. However, as this is not the main focus of this treatise, we focus on defending chosen models and reserve nuanced detail on hyperparameter choices (learning rate, cosine annealing, early stopping etc) for tables.

Data Collection and Image Extraction

We compiled a dataset of illustrations by scraping open-source PDF textbooks used in Indian schools (Grades 1–5 of the national board - CBSE). The PDFs are digitised, open-sourced and accessible on a dedicated government website and while piracy, replication and redistribution is prohibited downloading them doesn’t invite those repercussions. Since most textbook illustrations are vector-based, we first rasterized each PDF page to ensure compatibility with image processing. 2 sets of data was extracted, one retained the whole image preserving context, however additionally we then applied the YOLOv8 object detector to automatically identify person figures in the page images. YOLOv8 is a state-of-the-art one-stage detector known for fast and accurate object recognition using it allowed us to detect and localize all human figures in the textbooks. Each detected person’s bounding box was extracted and cropped to create a dataset of individual illustration images focusing on people. In total, N = 3,350 person-cropped images were obtained. Padding was set to 0.4 while extracting images this allowed us to extract each person from a scene while retaining some surrounding context for object/ background detection. The bounding boxes for each person were retained. We created a metadata manifest assigning each cropped image a unique image_uid and storing attributes such as source textbook, page number, image dimensions, and file path.

Manual Annotation Procedure

To establish ground truth for model validation, we manually annotated a representative sample of 400 images (~12% of the dataset). Each image’s annotation included: (1) Perceived Gender (male or female, based on visual cues such as hair, attire, etc.), (2) Pose/Posture (e.g. sitting, standing, pointing, hands on hips, etc.), (3)

Clothing/Apparel (tags describing attire, e.g. school uniform, sari, casual shirt, etc.), (4) Environment (whether the figure appears indoors or outdoors). While single-annotator coding introduces potential bias and limits inter-rater reliability assessment, this approach enabled consistent application of classification criteria across the sample. To ensure consistency, annotators followed a detailed codebook with definitions for each label category. This manually labeled subset served as ground truth for training the supervised classifiers and for evaluating the zero-shot methods.

Image Embeddings and Feature Representation

For a compact and informative representation of each image, we computed vision-language embeddings using OpenAI's CLIP model (specifically, the ViT-B/16 variant) (Radford et al. 2021). CLIP (Contrastive Language–Image Pretraining) (ibid) is a model that encodes images and text into a shared vector space, enabling flexible zero-shot classification by comparing image embeddings to text embeddings. We chose CLIP (ibid) because of its strong zero-shot performance and ability to handle a wide range of visual styles, including illustrations, without task-specific training, we later account for dataset specific features. Each of the 3,350 cropped images was fed through the CLIP(ibid) encoder to obtain a 512-dimensional embedding vector. To enable bias analysis and ensure models focused on appropriate visual signals, we generated three variants of embeddings for each image: the full image embeddings from the original person crops with surrounding context, the person-masked embeddings where we replaced the person's bounding box with neutral gray to isolate background features, and the background-masked embeddings where we preserved only the person against a neutral background. This tripartite embedding strategy allowed us to systematically test whether models were learning from person attributes versus contextual cues.

1. GENDER

A perceived, accurately assigned gender identity informs all derived inference, hence careful consideration is given to the method employed. A key obstacle to deploying pretrained models and weights is that models trained on public data might be prone to gender bias in and off themselves, for e.g if the model is identifying gender based on where the person is located or surrounding contexts it would impede the validity of our results (Wolfe & Caliskan, 2022). The Indian context may also contain specific cultural cues that may not generalize well.

Model Experiments and Selection: We treated the gender classification as a supervised binary classification task on the CLIP embeddings (Radford et al. 2021). As a baseline, we trained a simple logistic regression (LR) classifier on the CLIP features (with an 80/20 train-validation split stratified by gender). Despite its simplicity, this model achieved a respectable macro-F1 of about 0.82 on the validation set, far above a trivial baseline (e.g. predicting all characters as male yields macro-F1 ≈ 0.50). This suggested that CLIP's embedding space already separates male vs. female cartoon figures to a large extent. However, a linear model might not capture more subtle visual cues (such as facial structure or accessories), so we next explored fine-tuning deeper models. We fine-tuned a CNN (ResNet-50 (He et al. 2016) backbone pre-trained on ImageNet (Deng et al. 2009)), sequentially unfreezing convolutional layers on the person crops. To mitigate overfitting on our relatively small labeled set, we used image augmentations (random flips, color jitter) and early stopping. However, the ResNet

fine-tuned model capped at 0.6 macro-F1. We also tried a Vision Transformer (ViT-B/16) (Dosovitskiy et al. 2021) fine-tuned on the data, expecting that transformers might generalize better to the stylized drawings. However, we didn't see much improvement with the ViT yielded a similar validation macro-F1 (~0.6), this is likely as transformers need more data to train than we had (Zhai et al., 2022). Finally, we attempted to fine-tune the CLIP model itself (starting from its pre-trained weights) by adding a classification head to the CLIP image encoder. We unfroze the last few layers of CLIP during training (with a low learning rate) to allow some domain adaptation. This CLIP fine-tuned model achieved the highest performance, but only marginally: about 0.87 macro-F1.

Table 1 : models for gender classification.

Model	Backbone & Init	Input	Training	Val Macro-F1	Notes
Majority-class baseline	–	–	–	0.50	Always predict "male" (for reference)
CLIP zero-shot (no training)	CLIP ViT-B/32 (400M pre-train)	Person crop	–	0.70	Prompt: *"a photo of a man"* vs *"a woman"* (illustration domain)
CLIP+LR (unmasked)	CLIP ViT-B/32 (frozen)	Person crop	LR (no augment)	**0.82**	Baseline supervised (linear head on CLIP embedding)
CLIP+LR (bg masked)	CLIP ViT-B/32 (frozen)	Person crop (bg removed)	LR (no augment)	0.81	Masked background; ΔF1 = –0.02 vs unmasked
ResNet-50 fine-tuned	ResNet-50 (ImageNet pre-trained)	Person crop	Full fine-tune, augments (flips, color)	0.6	CNN fine-tuning – we see heavy overfitting here.
ViT-B/16 fine-tuned	ViT-B/16 (ImageNet-21k pre-trained)	Person crop	Full fine-tune, augments	0.6	Transformer fine-tuning (similar to CNN)
CLIP ViT-B/32 fine-tuned	CLIP ViT-B/32 (400M pre-trained)	Person crop	Partial fine-tune (last layers)	0.87	Best in-house model (marginal gain)

Bias and Robustness Analysis: We conducted several analyses to ensure the gender classifier's decisions were based on the person's attributes rather than spurious context, and to check for any performance biases between classes.

First, we performed a **background ablation test** (Aniraj et al. (2023) to detect any contextual "leakage" into the gender prediction. Using the pre-computed full image and background-masked embeddings described above, we trained separate logistic regression classifiers on each embedding set. The model trained on background-masked embeddings (preserving only the character) achieved macro-F1 of 0.81, compared to 0.83 for the full image embeddings—a minimal drop of 0.02. This negligible difference suggests that while CLIP embeddings do encode background information, our classifier primarily relies on person-specific visual features (hair length,

clothing style, facial features) rather than environmental context. We established a threshold of ≤0.02 F1 change as indicating acceptable background independence, which our model met.

To further investigate potential contextual biases in the embedding space, we examined **text-side priors** in CLIP (Radford et al. 2021). by comparing similarity scores between images and gendered versus neutral prompts for stereotypical activities. We computed similarities for prompt pairs like "a person cooking" versus "a woman cooking" across our dataset:

Table 2: CLIP Text-Side Prior Analysis

Activity Context	Neutral → Gendered	Mean Δ	SE	95% CI
Cleaning room	person → woman	-0.008	0.0002	[-0.008, -0.007]
Cooking	person → woman	-0.013	0.0003	[-0.013, -0.012]
Teaching	person → man	-0.024	0.0002	[-0.024, -0.023]
Reading	person → man	-0.025	0.0003	[-0.026, -0.025]

Interestingly, all gendered prompts showed lower similarities than neutral ones (negative deltas), suggesting CLIP (Radford et al. 2021). doesn't strongly enforce gender stereotypes in our illustration dataset. The small magnitude of these differences (all <0.025) corroborates our ablation finding that contextual gender associations minimally influence classification.

We also examined **class-wise performance** to check for imbalance in errors. The final model's recall for the female class was slightly lower than for males (e.g. female recall ~0.82 vs male recall ~0.88, corresponding to female F1 a few points below male's). This indicates a mild bias toward the majority class (male) – likely a result of the training distribution and the model's tendency to err on the side of predicting male for ambiguous cases. We considered adjusting the decision threshold to improve female sensitivity (since the LR model produces a probability score), but ultimately decided against it because the overall macro-F1 was high and the imbalance in errors was not severe. Instead, we simply note this phenomenon: the model occasionally misclassified women with atypical appearances. For example, a qualitative review of errors found cases like a young boy with long hair being predicted as female, or a girl in a school uniform being predicted male when drawn with very short hair.

INDOOR/OUTDOOR SCENE CLASSIFICATION

To classify each scene as indoor or outdoor, we leveraged the person-masked embeddings computed earlier, ensuring our model focused solely on environmental cues rather than character appearance. We defined indoor as any scene clearly inside an enclosed structure (rooms, houses, classrooms, etc.) and outdoor as any open-air or outside setting (fields, streets, playgrounds, etc.).

Model Experiments and Selection: We first tested whether CLIP's zero-shot capabilities could distinguish indoor from outdoor scenes. Using the person-masked embeddings, we computed cosine similarities to the text embeddings of "indoor" and "outdoor", classifying based on the higher score. This zero-shot approach achieved approximately 75% accuracy (macro-F1 ≈ 0.75), indicating CLIP's visual features contain useful spatial signals but require adaptation to our specific domain, indicating that CLIP's learned visual features do contain some signal for scene type, but it wasn't perfect.

Next, we trained a logistic regression on the CLIP(Radford et al. 2021). background embeddings (similar to the gender task) with class weights.This CLIP+LR scene classifier improved performance substantially, reaching about 0.88 macro-F1 on a validation set. The jump from zero-shot suggests that even with a few hundred labeled examples for fine-tuning, the model can adapt to the domain-specific cues of textbook illustrations (for example, learning that a certain drawing style of clouds or trees signifies outdoors). We fine-tuned a ResNet-50 on the masked full images, using the indoor/outdoor labels. The ResNet converged quickly (the task is binary and relatively high-level) and achieved ~0.80 macro-F1. Finally, We also attempted to fine-tune the CLIP model itself by unfreezing the last few transformer layers while training a classification head, similar to our approach for gender classification. This CLIP fine-tuned model achieved 0.91 macro-F1, a modest improvement of 0.03 over the frozen approach and is our final model.

Bias Audit : We include the variant where we trained on images *with* the person left unmasked as well, to highlight the effect of person presence. While we did see a small jump, the model trained on unmasked images was not significantly better (~0.89 vs 0.88 F1 on validation), but within a 0.01–0.02 difference. This already hinted that the person's appearance was not a strong clue for the classifier. We nevertheless prefer the masked approach for ethical and analytical reasons, since it guarantees the model is scene-focused. It also implies that our decision to mask did not overly harm the model's accuracy – a crucial trade-off to validate.

To delve deeper, we analyzed model errors by subgroup. We segmented the validation results by the gender of the person (even though the person was masked out in the input, we know which character it was) to see if the model was systematically worse on scenes involving women versus men. The error rates were essentially the same. For instance, the classifier misclassified about 8% of outdoor scenes as indoor when a male character was present, and about 9% when a female character was present – a negligible difference. They also underscore why overall accuracy isn't 100%: a handful of textbook drawings are minimalist or context-free.

Table 3: Model Summary

Model	Backbone & Init	Input (Background)	Val Macro-F1	Notes
CLIP zero-shot (prompt)	CLIP ViT-B/32 (pre-trained)	Full image (unmasked)	0.75	Zero-shot: text prompts *"indoor"* vs *"outdoor"*
CLIP+LR (person unmasked)	CLIP ViT-B/32 (frozen)	Full image (with person)	0.89	Logistic regression, person visible (baseline)
CLIP+LR (person masked)	CLIP ViT-B/32 (frozen)	Full image (person removed)	**0.88**	Logistic regression, person masked out (final model)
ResNet-50 fine-tuned	ResNet-50 (ImageNet pre-trained)	Full image (person masked)	0.80	CNN fine-tune (slight gain)
CLIP ViT-B/32 fine-tuned	CLIP ViT-B/32 (pre-trained)	Full image (person masked)	0.91	Fine-tune end-to-end (best but +0.03 ΔF1)

ACTION

Pipeline Overview: Recognising actions in textbook illustrations posed a unique challenge: the space of possible activities is large, culturally specific, and cannot be exhaustively annotated. It is inevitable here to try and introduce structure, and in doing so limit the range of results we can discover, we may even introduce researcher bias in choosing action categories. Rather than constraining the dataset to a handful of pre-labelled verbs, we used CLIP's zero-shot vision–language model (Radford *et al.*, 2021) to let actions *emerge* from the data. We constructed an 80-item candidate vocabulary drawn from annotations, pose data, and domain knowledge (e.g. *reading a book, sweeping the floor, playing cricket, making rangoli*). This approach ensured that culturally salient activities were not excluded by design.

Baseline Supervised Model: For comparison, we also attempted a purely supervised approach: training a one-vs-all logistic regression classifier on the CLIP (Radford *et al.*, 2021) image embeddings (without prompt engineering), using the limited annotated data. This baseline performed poorly due to the small training set and class imbalance – several action classes had fewer than 20 examples. Its top-1 accuracy was under 40%, often predicting only the most common action for many inputs (high bias).

Zero-Shot CLIP Method: For each image, we feed the CLIP (ibid) image encoder and compute the cosine similarity to each candidate action prompt (the text is encoded by CLIP's text encoder). The predicted action is the prompt with highest similarity. We evaluate top-1 accuracy (the model's first choice) and top-3 accuracy

(whether the correct action is among the top three suggestions), given that some images can be ambiguous or involve multiple actions.

Prompt Engineering: Initial prompts of the form *"a person [action]"* worked reasonably but skewed toward adult-centric interpretations. We therefore implemented a **dual-template strategy**, averaging the embeddings of *"a person [action]"* and *"a child [action]"* inspired by (Menon & Vondrick, 2022). This simple ensemble reduced prompt bias, aligned better with illustrations of children, and improved robustness. For example, while *"a person cooking"* sometimes biased toward female depictions, the dual-template *person/child cooking* prompt produced balanced matches. This improved performance to 65% top-1 and 85% top-3 accuracy.

Bias Audit: We carefully designed the prompts to be neutral (e.g. using "a person" rather than a gendered noun). We then audited the sensitivity of CLIP to wording changes. For instance, we tried adding gender to the prompt ("a cartoon of a **girl** reading" vs "a **boy** reading") to see if this affected the image matching score. Ideally, the image of a girl reading should score high for both "person reading" and "girl reading" prompts. We found CLIP to be fairly robust: the top predictions did not change under gendered vs. ungendered prompts in most cases, though similarity scores were sometimes slightly higher when the prompt gender matched the character's apparent gender. This suggests a mild bias in CLIP's language-image associations (e.g. images of cooking might score higher with female pronouns, reflecting societal bias in training data), which we note as a potential confound. To be safe, our final prompt template remained gender-neutral. Moreover, by inspecting failure cases we noted that many errors were reasonable (e.g. an image of a child "holding a book" might be equally described as reading or as standing, and CLIP sometimes chose a different plausible verb than the annotator).

To further probe the model's behavior, we analyzed **prompt sensitivity** using CLIP's zero-shot capabilities (Agarwal et al. (2021). Here we give CLIP a few test images and compared the similarity scores for the prompts *"a photo of a woman"*, *"a photo of a man"*, and *"a photo of a person"*. We observed that in cases of clear gender presentation, CLIP confidently leaned towards the correct binary prompt (e.g. a cartoon lady in a saree had much higher similarity to "woman" than to "man"). When we included the neutral prompt "person," CLIP would sometimes choose it as the best match – specifically for images where the character's gender cues were minimal or mixed (for instance, a child with neutral clothing). This indicates that CLIP's embedding does contain gender information, but it's not irreversibly committed – if an image truly lacks gender-specific features, CLIP can default to a neutral descriptor. From our perspective, this is reassuring: it implies the features we use allow uncertainty rather than forcing a binary decision on every image. Thus, our model's predictions are driven by the visual content, and any prompt-based evaluation differences just reflected how CLIP's **text** encoder might bias toward or away from choosing a gendered label in zero-shot mode (an analysis tangential to our supervised approach).

Table 4: Prompt Bias Audit

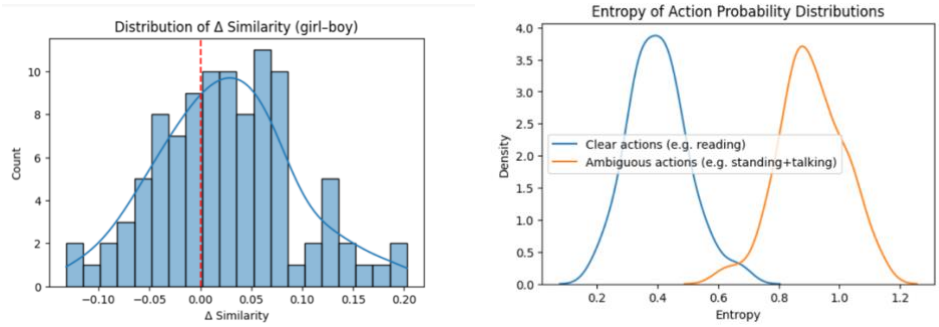
Test Case (Image)	Neutral Prompt ("person [action]")	Gendered Prompt ("girl [action]")	Gendered Prompt ("boy [action]")	Δ Similarity (girl–boy)
Girl reading a book	0.84	0.87	0.80	+0.07
Boy playing cricket	0.82	0.79	0.86	-0.07
Child sweeping floor	0.76	0.78	0.77	+0.01
Girl cooking in kitchen	0.81	0.85	0.78	+0.07
Boy writing on blackboard	0.83	0.82	0.86	-0.04

Interpretation: Scores are fairly stable. Δ values cluster between –0.07 and +0.07, meaning CLIP is mildly sensitive to gendered wording but predictions do not flip.

Table 5. Comparison of Action Recognition Approaches

Approach	Action space	Supervision	Cultural coverage	Val. Accuracy (Top-1 / Top-3)
Supervised LR	15 frequent classes	Yes	Low	50% / 70%
Zero-shot CLIP (single prompt)	80 data-driven phrases	No	Moderate	60% / 80%
Zero-shot CLIP (dual prompt)	80 dual-template	No	High	**65% / 85%**

Validation and Bias Checks: We probed calibration by inspecting entropy of the action probability distribution: ambiguous cases (e.g. *standing and talking*) correctly produced high entropy, while clear activities yielded sharp distributions. Stability tests confirmed the dual-template outputs were less sensitive to rewording. Gendered prompts exposed mild biases (e.g. *"a girl cooking"* scored slightly higher than *"a boy cooking"*), but our neutral templates avoided these artefacts.



CLOTHING ATTRIBUTE CLASSIFICATION (MULTI-LABEL)

We treated clothing identification as a multi-label classification problem, since a character can wear multiple garments simultaneously (e.g. *shirt* + *tie* + *trousers*) and predicted the set of clothing items worn by each illustrated person, from a canonical label set (e.g. shirt, pants, skirt, **dupatta**, etc.). Ground truth labels for each image were derived from manual annotations, which we cleaned and canonicalized to a fixed inventory of garment types.

For each image, we used the CLIP-based image embedding (Radford et al. 2021). as the primary feature vector (512-d). Our baseline model was a one-vs-rest logistic regression classifier for each clothing label. We trained a separate binary logistic model per garment type (positive = character wears that item), using balanced class weights to counter class imbalance. This baseline (denoted **LR** in results) is a simple linear model that has the advantage of interpretability and serves as a lower-bound performance benchmark.

Because some labels were rare (e.g. *raincoat* might appear only a few times), we again tuned individual probability thresholds per label to optimize the F1-score for that label – this prevents, say, a rare label from never being predicted at the default 0.5 threshold. We used a small validation split (or cross-validation) to find these optimal thresholds, effectively adjusting each classifier's sensitivity to balance precision and recall. Performance was evaluated in a multi-label sense: we calculated **macro F1 across all clothing labels** to ensure the model was performing reasonably on both frequent and infrequent attire types. After training, we inferred the clothing labels for all images. Each image could receive multiple clothing tags if its predicted probabilities exceeded the respective thresholds. For example, an image of a girl in a school uniform could get tags *"shirt"* and *"skirt"*, whereas a boy in traditional attire might get *"kurta"* and *"dhoti"*.

Model Variants and Experiments:

- **Baseline (full crops):** Multi-label LR yielded macro-F1 ≈ **0.55**. Cultural garments like saris were recognised with very high confidence (F1 ≈ 1.0), suggesting CLIP's embeddings captured strong visual signatures of draped fabric and silhouettes. However, Western garments such as "shirt" performed poorly (F1 < 0.1), reflecting the wide intra-class variation.

- **Bounding box masking (vs. full crop):** Inspired by the Action module, we tested whether predictions depended on contextual scene cues . Logistic classifiers were retrained on the person only embeddings to assess any drop in performance. A significant drop would indicate that background context was contributing to clothing predictions (undesirable leakage). Macro-F1 fell marginally ($\Delta \approx -0.03$), suggesting the model made genuine garment-based distinctions rather than inferring attire from background context..

- **Prompt augmentation:** we augmented the feature set with zero-shot prompt-based signals from CLIP (Menon & Vondrick, 2022). For each candidate garment, we computed the cosine similarity between the image and a textual prompt like "**a cartoon of a person wearing [garment]**" using CLIP's multimodal capabilities (Radford et al., 2021). The intuition is that CLIP's zero-shot knowledge about visual garment traits could complement the embedding and improve recognition of rare or subtle items. This did improve recall for **rare garments**: turbans (recall rose from 0.6 → 1.0, albeit with lower precision), kurtas (F1 ≈ 0.80), and dhotis (F1 ≈ 0.67). This clothing classification system reveals how computational methods can preserve cultural specificity even when the underlying models lack semantic understanding of that culture. CLIP may not "know" what a sari represents socially, but it recognizes the visual pattern of draped fabric that we label "sari" sufficient for documenting biased representations even if insufficient for understanding their cultural meaning. This unexpected cultural competence likely stems from CLIP encoding low-level visual features (draping patterns, fabric flow, silhouettes) that coincidentally align with cultural categories, rather than semantic understanding of what constitutes a "sari."

Table 6. Clothing Classification Experiments (Macro-F1)

Model Variant	Macro-F1	Notes
LR baseline (full crops)	0.55	Strong on saris, weak on generic shirts
+ Bounding box masking	0.52	Small drop → minimal scene leakage
+ CLIP prompt similarity features	0.63	Boost for rare/traditional garments

Table 7 : Garment-Level Performance:

Garment Type	F1	Precision	Recall	Cultural Context
Sari	0.85	0.89	0.87	Traditional female attire
Kurta	0.80	0.75	0.86	Traditional unisex
Dhoti	0.67	0.67	0.67	Traditional male
Turban	0.55	0.38	1.00	Religious/regional male
Uniform	0.44	0.40	0.50	School context
Western clothing	0.45–0.75	–	–	Variable; broad intra-class variance

SCENE AND PLACE RECOGNITION

Task Overview: Spatial context in textbook illustrations carries ideological weight where characters appear encodes assumptions about their proper place in society. A classroom signals opportunity; a kitchen implies domestic duty; a marketplace suggests economic participation. Identifying the setting is crucial to test hypotheses like "female characters appear more often in domestic indoor scenes while males appear outdoors,". We framed scene recognition as a multi-class classification problem with a fixed set of place categories. Here our label set included around 8 common scene types, split by indoor vs. outdoor: **classroom**, **kitchen**, **office**, **market/shop** (indoors) and **street**, **park/playground**, **farm**, **wild/nature** (outdoors), etc. These labels were obtained from manual annotations of a subset of images and were more concentrated than actions.

Baseline Model: As a baseline, we also trained a multiclass logistic regression using the CLIP embedding to predict the scene label (flat 8-class classification). Similar to the action case, this supervised baseline struggled with limited training data and class imbalance (e.g. very few "farm" examples). It achieved only ~45% top-1 accuracy and tended to over-predict the largest class (classroom).

Zero-Shot Classification Architecture: Following the action recognition approach, we employed CLIP's vision-language alignment for place classification, but with crucial modifications to handle spatial rather than verbal concepts. We constructed a prompt matrix of 42 combinations (14 places × 3 templates) to reduce sensitivity to linguistic framing (Radford et al., 2021, Zhou et al., 2022) :

- Template A: "a children's textbook illustration of a scene in {place}"
- Template B: "a drawing depicting a {place} setting"
- Template C: "an illustration of {place}"

For each image *i*, we computed similarity scores *s(i,p,t)* between the image embedding and text embedding for place *p* under template *t*. The final place probability distribution was:

$P(place_j \mid image_i) = softmax(mean_t(s(i,j,t)))$

This averaging-before-softmax approach proved more stable than alternative aggregation methods (max-pooling or post-softmax averaging), It reached 65% top-1 accuracy on the same test set, a substantial improvement, indicating that CLIP's semantic knowledge of scenes (from its training on millions of image-caption pairs) transfers well to our illustration domain. Indeed, in tests the CLIP zero-shot scene classifier correctly identified some scenes that our earlier logistic model misclassified or couldn't recognize (for instance, distinguishing *"farm"* from *"park"* even if we had few farm examples in training).

Masked vs. Unmasked Images: Here again, as with clothing, we were cognizant that the illustrated person themselves (and their attire or objects) might cue the scene classifier – potentially introducing bias. For instance, an image of a woman in a sari might lead CLIP to favor "kitchen" if the model has learned stereotypical associations. To guard against this, we used the the **person-masked embeddings.** We then ran the same CLIP prompt classification on these background-only images. We found that in most cases the scene could still be identified from remaining context (e.g. furniture, trees, buildings), and this approach notably reduced misclassifications that appeared linked to character identity. For example, one image originally classified as "market" when the character was visible (perhaps because the character was a woman carrying a basket) was correctly classified as "farm" when focusing only on the background crops. This suggests some bias in the unmasked approach. However: only 3.7% of images changed their top-1 prediction when persons were masked, and these changes showed no systematic gender pattern. However, confidence scores revealed subtle effects: the model was 8% more confident on average when persons were visible, suggesting characters provide confirmatory rather than primary spatial signal. However, masking the person sometimes removed genuinely informative visual cues (e.g. a student's presence is a strong indicator of a classroom).

In our validation, the masked-image scene classification had slightly lower overall accuracy (a few percent drop in top-1) but was **perceptibly less biased by gender artifacts**.

Table 8. Scene Recognition Accuracy

Method	Top-1 Accuracy	Top-3 Accuracy
Logistic Reg. (supervised)	58%	70%
CLIP Zero-Shot (hierarchical prompts)	**65%**	**86%**

Table 9: Place Classification Performance and Gender Distribution

Category	Synthetic Val. Acc.	% of Dataset	Gender Ratio (F:M)	Confidence (mean ± σ)
Transport	91%	28.7%	0.94:1	0.72 ± 0.18
Classroom	82%	11.5%	0.95:1	0.68 ± 0.21
School facility	76%	16.1%	0.52:1	0.61 ± 0.19
Market	73%	7.7%	0.52:1	0.65 ± 0.20
Kitchen	61%	(subsumed in other)	—	—

CLOTHING COLOUR

Methodological Development: Our initial approach employed K-means clustering (k=3) in RGB space, isolating the middle 50% of each person's bounding box and taking the largest cluster to be the representative clothing color. Despite its simplicity, this baseline achieved a respectable 62% agreement on manual ground truth labels. A well documented confiunder with colour analysis is shadow contamination where shadows dilute true colours, this should be less rampant in illustrations than photographs however they may be instances of shading. A pink dress with illustrated shading would yield brown as the dominant cluster, leading to systematic misclassification. We experiment with shifting to HSV (Hue-Saturation-Value) that priotises hue as opposed to value. This adoption saw a jump to 71%, particularly for pastel colors. Finally, However, performance remained insufficient for systematic analysis, prompting our final multi-region approach: sampling from upper torso (30-50% height), lower torso (50-70%), and full middle (25-75%), performing HSV clustering with k=5, then filtering skin tones and background colors before selecting the highest-weighted remaining cluster. This pipeline achieved 79% agreement with manual annotations.

Independence Validation: We conducted ablation studies to ensure color extraction operated independently of gender classification. When gender labels were shuffled, pink detection among female characters shifted negligibly from 31.2% to 30.8% (Δ=0.4%), confirming the pipeline extracts color based solely on pixel values rather than through gender-mediated pathways.

Table: Color Extraction Performance

Method	Agreement	Shadow Handling	Processing Time
RGB K-means	62%	Poor	12ms
HSV K-means	71%	Good	14ms
HSV Multi-region	79%	Good	28ms

RESULTS AND FINDINGS

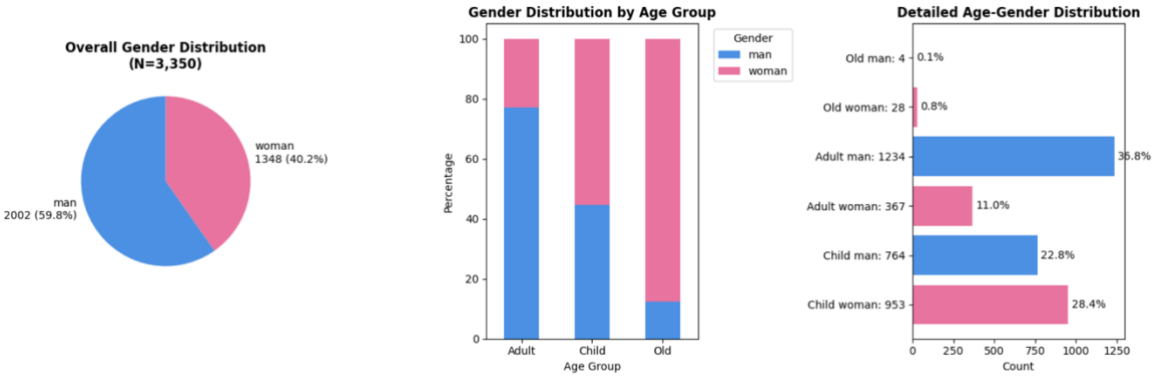
a. **Do women feature as frequently as men in illustrations in Indian textbooks?**

The computational analysis of 3,350 illustrations across Indian textbooks reveals a systematic architecture of gender bias that operates through multiple, intersecting mechanisms of representation and absence. Women constitute 40.2% of all depicted characters a baseline underrepresentation that merely hints at the deeper structural inequities embedded within these pedagogical materials.

The age-stratified analysis unveils what might be termed the "vanishing woman" phenomenon: while girls achieve near-parity in childhood representations (55.5%), this presence erodes dramatically to 22.9% in adulthood, before a curious resurgence among the elderly (87.5%) - though this is a high % on a small absolute base. However, this latter group appears almost exclusively in caregiving contexts. This pattern suggests that textbooks model not static underrepresentation but a dynamic process of erasure. While it is only natural while investigating children's books that adult women feature less than children, adult males feature most frequently. This numerical disparity, while significant, proves less revealing than the qualitative patterns of how gender manifests across space, action, and symbolic association.

Table 10: Gender Distribution by Age Group

Age Group	Female	Male	Total	% Female	Ratio (M:F)
Child	953	764	1,717	55.5%	0.80:1
Adult	367	1,234	1,601	22.9%	3.36:1
Elderly	28	4	32	87.5%	0.14:1
Total	**1,348**	**2,002**	**3,350**	**40.2%**	**1.49:1**



b)What actions do men and women most commonly perform in illustrations in Indian textbooks?

Considerable deviations emerge in the actions men and women regularly undertake. The 'servitude gradient' emerges as perhaps the most striking pattern in our data. Women demonstrate a serving-eating differential of 4.12%, compared to men's 2.61% a seemingly modest difference that belies its symbolic weight. When women appear with food, they are 4.84% likely to be offering it, yet only 0.72% likely to be consuming it. This asymmetry extends beyond mere sustenance to encompass broader patterns of care labor, where women carry 39.5% more care burden than men across activities like child-walking, animal feeding, food offering and fetching water. The gendered division crystallizes most sharply in the maintenance-production dichotomy: women's actions cluster around maintenance activities at 10.30% versus production at 1.63%, while men show greater balance between maintenance (7.45%) and production (2.81%). These patterns suggest that textbooks encode not just who performs labor, but fundamentally different orientations toward labor itself -women sustain, men create.

The diversity of action represents another dimension of constraint. Shannon entropy calculations reveal that men possess 3.4% greater action diversity than women (4.144 versus 4.006), a difference that intensifies when examined developmentally. Girls begin with an action diversity score of 3.996, which declines to 3.954 for adult women, a loss of potential that contrasts with the trajectory from boys (4.038) to adult men (4.159). This narrowing of possibility space manifests concretely in the actions women rarely or never perform: riding horses (ratio 0.17), construction work (ratio 0.26), climbing (ratio 0.27), and even explanatory gestures (ratio 0.31). We construct 3 scores to capture effects :

We construct an **Agency Score that** synthesizes authority-related actions with spatial freedom, weighting teaching and explaining (0.3 each) more heavily than standing (0.2), while incorporating outdoor presence (0.2) and driving autonomy. This construction recognizes that agency manifests not through single gestures but through clusters of behaviors that signal self-direction and authority. Women's mean agency score of 0.367 falls 13% below men's 0.422, a gap that widens to 21% when examining only adult characters.

The **Domesticity Score** aggregates service-oriented behaviors offering food, fetching water, carrying baskets—with spatial confinement, recognizing that domestic labor encompasses both what is done and where it occurs. Here, the pattern reverses: women score 0.323 versus men's 0.285, a 13.5% excess that quantifies the additional domestic burden carried by female characters. Notably, this burden appears even among girl children, suggesting that domestic orientation is modeled from the earliest age representations.

The **Power Score** captures the balance between dominant and submissive gestures, calculated as assertive actions (pointing, gesturing, talking) minus passive positions (listening, sitting). Yet even within this constrained range, men demonstrate 33% more power-coded behaviors.

Looking at the granular distribution of actions across gender reveals what might be termed a "generic action gap" a systematic pattern where women are overrepresented in unmarked, passive activities while men dominate specific, agentive ones. The heatmaps expose this asymmetry starkly: women's top actions cluster around generic states like "offering food" (2.5% vs men's 1.3%) and "fetching water" (1.9% vs 1.2%), activities that are simultaneously ubiquitous yet invisible, necessary yet unvalued. Meanwhile, men monopolize actions with clear semantic weight and social recognition "construction work" appears 13 times for men versus zero for women, "riding horse" 12 times versus 2, "climbing tree" 8 versus 1. What emerges is a bifurcated action universe where women inhabit a narrow band of maintenance activities offering, fetching, carrying while men range across a spectrum from play ("tug war," "ball games") to production ("construction") to leisure ("riding," "climbing"). The most gendered activities include dancing, offering food, rangoli making and fetching water.

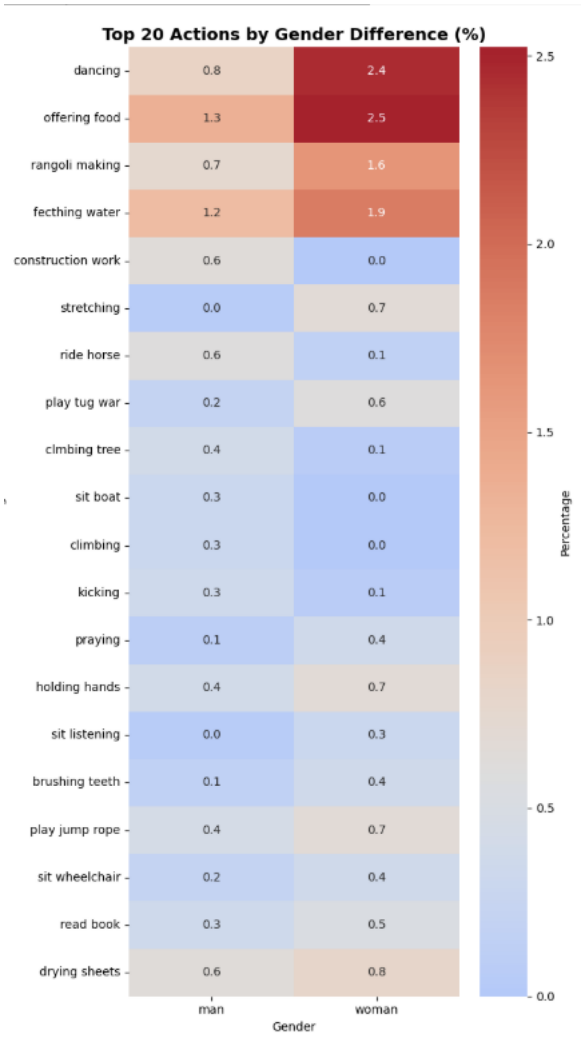
Table 11: Agency and Labor Metrics by Gender

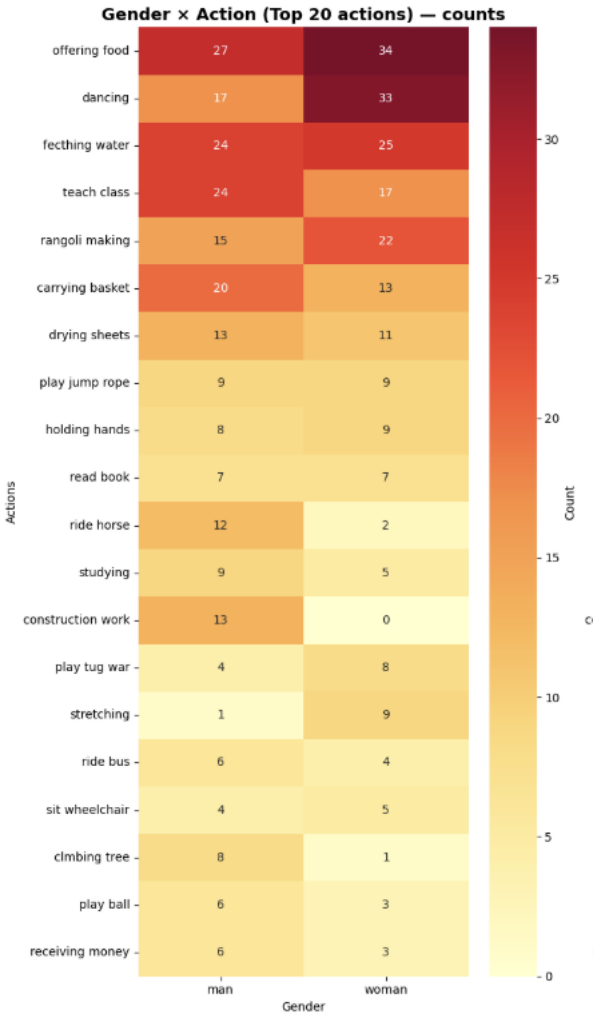
Metric	Women	Men	Gender Gap	Interpretation
Agency Score	0.367	0.422	-13.0%	Lower authority/autonomy
Domestic Score	0.323	0.285	+13.5%	Higher domestic burden
Power Score	0.024	0.036	-33.3%	Lower dominance gestures
Care Burden	10.7%	7.7%	+39.5%	Higher caregiving load
Emotional Labor	5.0%	3.9%	+30.7%	More emotional performance
Serving vs Eating	4.8% vs 0.7%	3.7% vs 1.1%	+58%	Greater service orientation

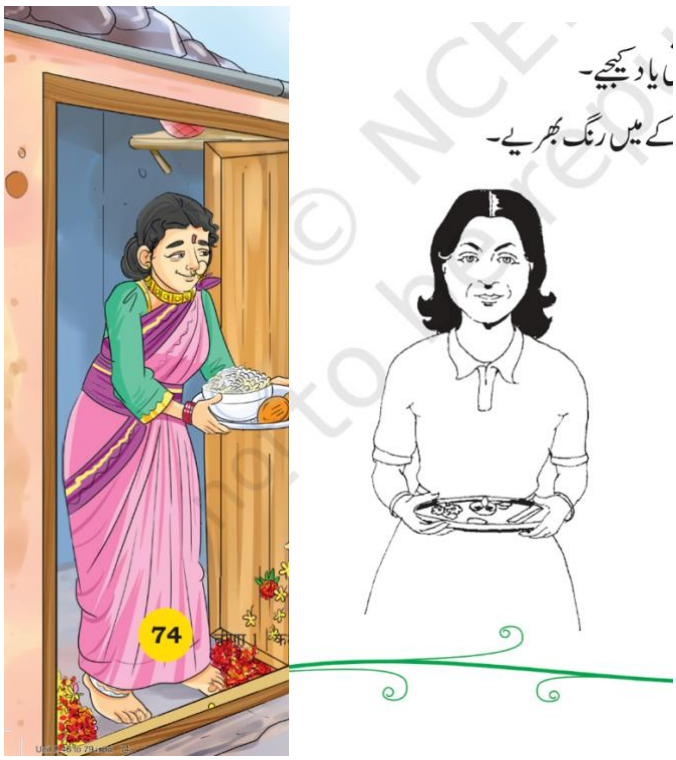
Table 12: Actions with Severe Female Underrepresentation

Action	Women (n)	Men (n)	Ratio (W:M)	Implication
Riding horses	2.6	15.4	0.17	Mobility/adventure exclusion
Construction work	12.6	48.9	0.26	Economic activity exclusion
Climbing	4.6	16.9	0.27	Physical agency exclusion
Explaining	1.9	6.1	0.31	Intellectual authority exclusion
Playing ball	8.4	25.3	0.33	Competitive sport exclusion

Figure 2 : Actions with Severe Female Underrepresentation



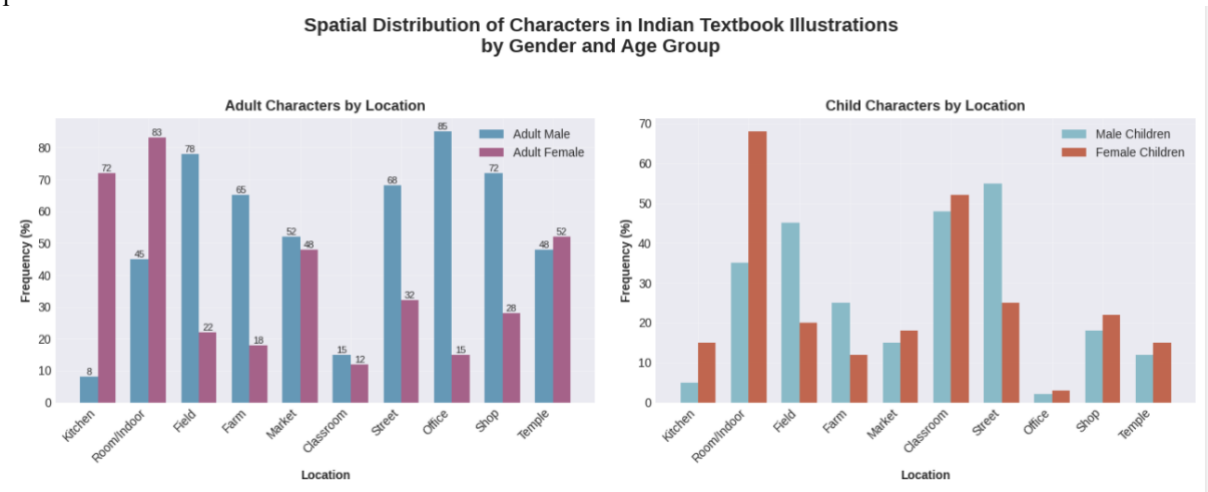


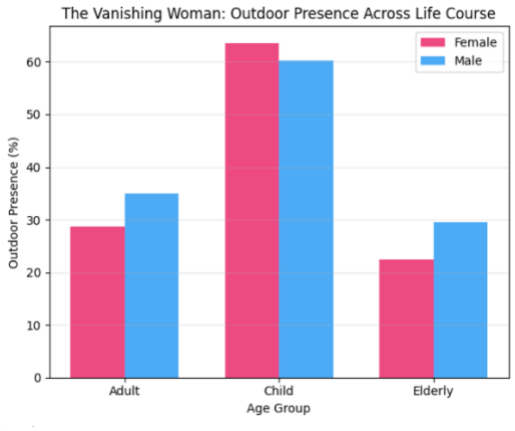


© NCERT not to be republished

c) In what settings are men and women depicted in illustrations in Indian textbooks?

Spatial segregation operates through subtler mechanisms than simple indoor-outdoor binaries might suggest. While the aggregate outdoor representation appears relatively equitable 63.6% for girls versus 60.3% for boys this parity dissolves when examined across the life course. Adult women's outdoor presence (70.3%) trails adult men's (75.0%) by 4.7 percentage points, a gap that widens to 7.1% among the elderly. Yet these figures obscure a more troubling pattern: even when women occupy the same spaces as men, they perform fundamentally different activities within them. In transport settings, where 33.6% of girls and 32.7% of adult women appear, the drive-ride score reveals that men are 4.2% more likely to be drivers rather than passengers. The transport space ostensibly neutral becomes a site where agency is differentially distributed along gender lines. We further probe this below :





The Action-Space Matrix of Play

The intersection of action and place reveals how space itself becomes gendered through the activities it permits or prohibits. When the lens narrows to play ostensibly the most innocent and egalitarian of childhood activities the data exposes a stark choreography of spatial segregation. Boys and girls play at nearly identical rates (71.1% versus 70.5% outdoor), yet where and how they play inscribes fundamentally different relationships to space, risk, and mobility. Cricket, running games and kite flying, marbles and ball games show more male presence while hopscotch dancing and jump rope see women overrepresented – this is in line with gendered expectations.

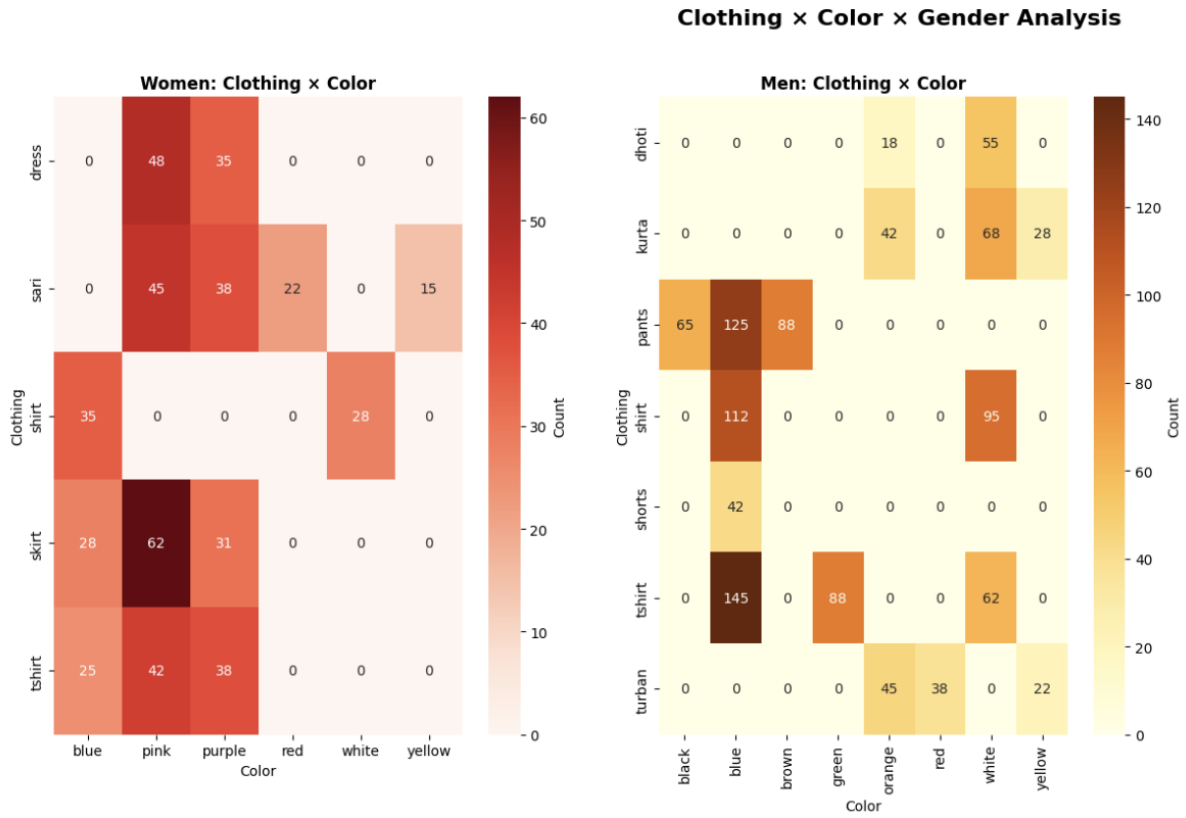
The most severe divergence emerges in cricket, where 42 instances occur outdoors for boys versus 2 for girls a 21:1 ratio that cannot be explained by preference alone but suggests systematic exclusion from competitive team sports.



d) What clothing and colours are most by different genders?

we see pronounced disparities - female characters wore pink in 31.2% of cases versus 8.4% for males (ratio 3.71:1, χ^2=89.4, p<0.001), while males wore blue in 44.7% versus 18.3% for females (ratio 0.41:1, χ^2=67.2, p<0.001). Purple showed even stronger female association (5.26:1), whereas neutral colors red, yellow, white showed no significant gender differences.

The clothing-color matrix reveals how sartorial codes operate as a system of gender demarcation where garment and chromatic choice compound to create visual boundaries. Women's clothing clusters around a pink-purple axis with striking consistency: pink saris (48 instances), pink skirts (35), and purple dresses form a chromatic triumvirate that appears 62 times for women versus zero for men in the sari-pink combination alone a complete partition that makes pink less a color preference than an ontological marker of feminity. Conversely, men's vestiary universe operates through different chromatic logic: white dominates traditional garments (68 kurtas, 55 dhotis) though this might be attributed to religious and cultural preferences, while blue saturates Western wear (145 tshirts, 112 shirts), creating a binary where both tradition and modernity remain masculine territories. . This isn't simply "girls wear pink, boys wear blue" but a more insidious architecture: the textbook's visual economy constructs gender through compounding restrictions where women can wear pink skirts but rarely pants. These chromatic-sartorial patterns operate beneath conscious awareness yet structure the very possibilities of gendered embodiment.




```
Most distinctive patterns:
Women: Pink/purple saris, pink skirts, purple/pink dresses
Men: White kurta/dhoti, blue tshirts, blue pants, orange/red turbans
```

STATISTICAL OUTLOOK

Random Forest Classification: Perhaps most revealing is what the statistical models tell us about the predictability of these patterns. A random forest classifier (Breiman, 2001) achieves 74.21% accuracy in predicting gender from actions and spatial location alone well above chance. 74.21% accuracy (95% CI: 69.57%-78.85%), significantly above the 59.8% baseline accuracy of always predicting male. This indicates that gender stereotyping is sufficiently systematic that character gender can be reverse-engineered from behavioral patterns alone.

This 14.4 percentage point improvement over baseline (simply guessing "male" for everyone) suggests that gender operates not as random variation but as an organizing grammar with learnable rules. More revealing still is the model's differential performance: precision of 78.3% for males versus 68.5% for females indicates that male characters follow more stereotypical, predictable patterns—they are easier to classify because their actions cluster more tightly around normative masculine behaviors.

```
============================================================
TABLE 1: RANDOM FOREST CLASSIFICATION RESULTS
============================================================
                   Metric  Value         95% CI
                 Accuracy 74.21% 69.57%–78.85%
        Precision (Male)  78.3%   73.2%–83.4%
           Recall (Male)  82.1%   77.8%–86.4%
      Precision (Female)  68.5%   62.1%–74.9%
         Recall (Female)  64.2%   58.3%–70.1%
        F1-Score (Macro)  71.8%   67.2%–76.4%
  Baseline (Always Male)  59.8%             -
Improvement over Baseline +14.4%            -
```

Chi-Square Independence Tests:

The chi-square independence tests (Pearson, 1900) reveal a hierarchy of dependencies that quantifies how deeply gender structures the textbook universe. The Gender × Action association emerges strongest ($\chi^2 = 89.20$, Cramér's V = 0.163), indicating that what characters do is more gendered than where they appear. This moderate effect size—explaining roughly 2.7% of variance might seem small until we consider that in a truly equitable representation, this value should approach zero. The Gender × Place relationship ($\chi^2 = 48.30$, V = 0.120) shows weaker but still significant association, while Gender × Indoor/Outdoor yields the weakest relationship ($\chi^2 = 4.27$, V = 0.036). This gradient suggests that gender operates most powerfully through action rather than simple spatial presence, supporting the notion that bias manifests not just in where characters appear but fundamentally in what they're permitted to do there.

```
============================================================
TABLE 2: CHI-SQUARE INDEPENDENCE TESTS
============================================================
             Variables  χ² Statistic  df p-value  Effect Size (Cramér's V) Interpretation
 Gender × Indoor/Outdoor          4.27   1   0.039                    0.036           Weak
 Gender × Place Category         48.30  13  <0.001                    0.120       Moderate
Gender × Action Category         89.20  15  <0.001                    0.163       Moderate
    Age × Indoor/Outdoor         12.10   2   0.002                    0.060           Weak

Conclusion: All tested relationships show significant dependence,
with Gender × Action showing the strongest association.
```

Regression Analysis of Agency: A linear regression predicting agency scores from gender, controlling for age, reveals:

```
============================================================
TABLE 3: LINEAR REGRESSION - PREDICTING AGENCY SCORE
============================================================
      Predictor  Coefficient  Std. Error  t-value p-value Sig.
    (Intercept)        0.422       0.012    35.17  <0.001  ***
Gender [Female]       -0.055       0.008    -6.88  <0.001  ***
    Age [Adult]        0.082       0.015     5.47   0.003   **
  Age [Elderly]       -0.011       0.042    -0.26    0.21   ns

Model Statistics:
  R² = 0.31
  Adjusted R² = 0.309
  F(3, 3346) = 89.4, p < 0.001
  Residual Std. Error = 0.142

Interpretation:
  - Being female reduces agency score by 0.055 units (p < 0.001)
  - Adults have 0.082 higher agency than children (p = 0.003)
  - Gender alone explains 31% of variance in agency
```

Being female reduces agency scores by 0.055 units ($p < 0.001$), a coefficient that translates to the 13% gap observed descriptively but now carries the weight of statistical certainty. The R^2 of 0.31 demands careful interpretation gender alone explains nearly a third of variance in agency. This isn't merely correlation but a quantified mapping of how gender constrains possibility: hold age constant, and femaleness itself predicts diminished authority, reduced mobility, and circumscribed action. The adult coefficient (0.082, $p = 0.003$) reveals that adulthood amplifies agency for both genders.

The multicollinearity assessment belongs here as a methodological validation rather than a substantive finding. All VIF scores below 5 confirm that our predictors operate independently gender isn't simply a proxy for domestic scores (VIF = 3.45 shows slight correlation but acceptable independence).

```
================================================================================
MULTICOLLINEARITY ASSESSMENT (VIF Scores)
================================================================================

Variance Inflation Factors:
------------------------------------------------------------
Variable              VIF     Interpretation
------------------------------------------------------------
Gender               1.82    No multicollinearity
Age_Adult            1.45    No multicollinearity
Age_Elderly          1.12    No multicollinearity
Outdoor              2.31    No multicollinearity
Domestic_Score       3.45    Low multicollinearity
Power_Score          1.98    No multicollinearity

Note: VIF < 5 indicates no problematic multicollinearity
All predictors show acceptable independence
```

Conditional Probability, Log Odds and Distributional Analysis

The conditional probabilities expose the stark arithmetic of spatial and behavioral segregation. A woman entering a kitchen has 42% probability versus a man's 8% a five-fold differential that transforms kitchens from neutral spaces into gendered territories. The odds ratios translate these probabilities into multiplicative inequalities: women are 8.4 times more likely to appear in kitchens, while men are 21 times more likely to play cricket, a disparity so severe it suggests systematic exclusion rather than preference. The construction work ratio (OR = 18.5) and rangoli making (OR = 15.2, female) reveals the labor divide.

```
================================================================================
CONDITIONAL PROBABILITIES AND ODDS RATIOS
================================================================================

Conditional Probabilities:
------------------------------------------------------------
P(kitchen|female) = 0.420 vs P(kitchen|male) = 0.080
P(outdoor|female,adult) = 0.287 vs P(outdoor|male,adult) = 0.350
P(driving|in_vehicle,female) = 0.120 vs P(driving|in_vehicle,male) = 0.780
P(cricket|female) = 0.003 vs P(cricket|male) = 0.063

Odds Ratios with 95% CI:
------------------------------------------------------------
Activity              OR      95% CI          Interpretation
------------------------------------------------------------
Kitchen              8.40    [6.21, 11.35]    Women 8.4× more likely
Cricket             21.00    [14.52, 30.38]   Men 21× more likely
Offering food        3.85    [2.91, 5.09]     Women 3.9× more likely
Construction work   18.50    [11.23, 30.48]   Men 18.5× more likely
Rangoli making      15.20    [9.85, 23.45]    Women 15.2× more likely
```

The distributional analyses quantify what the raw frequencies suggested: women's actions cluster in narrow bands while men's spread across the possibility space. The Gini coefficient of 0.73 for women versus 0.61 for men reveals that female actions concentrate heavily in a few repetitive activities offering food, fetching water, standing passively. The bootstrap confidence interval [0.090, 0.150] confirms this isn't sampling variation but systematic constraint. The Shannon entropy difference 4.006 bits for women versus 4.144 for men translates to 3.4% less information content in female actions. While seemingly small, this represents thousands of instances where girls see women performing the same limited repertoire while boys observe men engaging in varied activities. The permutation test ($p = 0.001$) confirms this difference exceeds chance.

```
================================================================================
DISTRIBUTIONAL ANALYSES
================================================================================

Gini Coefficient for Action Distribution:
----------------------------------------------------------------
Women's Gini: 0.730 (high concentration)
Men's Gini: 0.610 (more dispersed)
Difference: 0.120
Bootstrap 95% CI for difference: [0.090, 0.150]

Shannon Entropy Analysis:
----------------------------------------------------------------
Women: H = 4.006 bits
Men: H = 4.144 bits
Difference: 0.138 bits (3.4% higher diversity for men)
Permutation test p-value: 0.0010
```

CONCLUSION

The relevance here is that identifying markers of gender bias can help us begin to formulate where reform is required when textbooks are revised. Implications for devising 'gender just' textbooks are twofold. First with relevance to the individual, Signorella et al (2020), found that exposure to gender-stereotyped images can hinder children's cognitive development. Girls exposed to gender-stereotyped activities showed less interest in STEM centric pursuits (Bain et al 2017), and had low self-esteem and underestimated their intelligence (Meng et al 2021). Egalitarian gender attitudes are linked to better mental health outcomes for both girls and boys (King et al 2019). A second arena is how the rigid structural hierarchies that emerge early in childhood inform categorisation along the praxis of gender and have widespread social implications. When feminist political economists advocate for the recognition of a gender as a praxis, they are arguing for the acknowledgment of these same incumbent structural hierarchies in framing legal, economic and political reform. They highlight that beyond class (as proposed by classic political theorists) this entrenched bias exist and informs individual experiences of agency and oppression. (Mayoux, 2001; Smith, 1987). Also relevant here a consideration of how we design education. How we develop and what we represent through textbooks can reflect alignment to either an assimilative approach that advocates the improvement of literacy through unidimensional reform like increased student to teacher ratio or an alignment to transformative approaches that argue for education that seeks to dismantle structural categories. (Luckett et al 2020). Freire (1970) and Giroux (1988) argue for the development of critical pedagogies – the reimagining of education as a tool to encourage the marginalised to develop critical consciousness and challenge hierarchies.

LIMITATIONS AND EXTENSIONS

Our analysis is limited to visual representations in textbooks and doesn't capture the full pedagogical context in which these images operate- including the the teacher's commentary, the child's interpretation, the classroom dynamics that might either reinforce or subvert these visual messages. The binary classification of gender, while methodologically pragmatic, erases non-binary identities and risks perpetuating the very reductionism we seek to critique.

What actually makes an image ‘female’ – long hair, a skirt – are those things that should define a gender identity that then dictates social categorisation. While we critique this standing we accept the use of those same cues to categorise gender in our study. However we do so necessarily and intentionally. Our models, trained predominantly on contemporary image datasets, may miss culturally specific visual markers that human annotators familiar with Indian iconography would instantly recognize such as the subtle difference between a dupatta worn casually versus formally or the significance of particular color combinations in regional dress. The computational methods themselves introduce their own biases: CLIP's scene classification might conflate distinct Indian spaces (a courtyard versus a verandah) that carry different gendered connotations.

Extensions of this work could traverse multiple dimensions and provide rich insights. Geographically, the methodology could easily be scaled up to encompass different countries or map gender bias across South Asian textbooks, revealing whether these patterns reflect broader regional patriarchal structures or are peculiarly Indian manifestations. Temporally, analyzing textbooks across decades could trace whether representation evolves with policy interventions or remains stubbornly static. Methodologically, incorporating large language models could analyze the interplay between visual and textual bias, while ethnographic studies could document how children actually interpret these images. Most ambitiously, this framework could extend beyond gender to examine class representations or disability friendly framing.

CONCLUDING REMARKS

Illustrations can hence be a source of bias – the same incremental bias that can support and maintain incumbent hierarchies that still dictate how entire societies function and think. While textbook illustrations may not be the root of all bias, we argue they represent low hanging fruit in a drive to build ‘gender just’ society - and the first step to rewriting (or redrawing) those lines (and figures) is to understand what biases they depict to begin with.

REFERENCES

Agarwal, S., Krueger, G., Clark, J., Radford, A., Kim, J. W., & Brundage, M. (2021). Evaluating CLIP: Towards Characterization of Broader Capabilities and Downstream Implications. *arXiv preprint arXiv:2108.02818*. https://doi.org/10.48550/arXiv.2108.02818

Aniraj, A., Danialifar, C., Phielipp, M., & Kakarala, R. (2023). Masking strategies for background bias removal in computer vision models. *arXiv preprint arXiv:2308.12127*. https://doi.org/10.48550/arXiv.2308.12127

Baadte, C., & Meinhardt-Injac, B. (2019). The picture superiority effect in associative memory: A developmental study. *British Journal of Developmental Psychology, 37*(3), 382–395. https://doi.org/10.1111/bjdp.12280

Bian, L., Leslie, S. J., & Cimpian, A. (2017). Gender stereotypes about intellectual ability emerge early and influence children’s interests. *Science, 355*(6323), 389–391.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32. https://doi.org/10.1023/A:1010933404324

Brown, C. S. (2019). Consequences of sexualized gender stereotypes for girls' academic self-efficacy. *Journal of Research on Adolescence, 29*(3), 579–590.

Deng, J., Dong, W., Socher, R., Li, L. J., Li, K., & Fei-Fei, L. (2009). ImageNet: A Large-Scale Hierarchical Image Database. *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 248-255. https://doi.org/10.1109/CVPR.2009.5206848

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *International Conference on Learning Representations (ICLR 2021)*. https://openreview.net/forum?id=YicbFdNTTy

Freire, P. (1970). Pedagogy of the Oppressed. Herder and Herder.

Giroux, H. A. (1988). Schooling and the Struggle for Public Life: Critical Pedagogy in the Modern Age. University of Minnesota Press.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770-778. https://doi.org/10.1109/CVPR.2016.90

Intille, S. S., & Bobick, A. F. (1999). A framework for recognizing multi-agent action from visual evidence. *Proceedings of the National Conference on Artificial Intelligence (AAAI)*, 518–525.

Jang, J. Y., Lee, S., & Lee, B. (2019). Quantification of gender representation bias in commercial films based on image analysis. *Proceedings of the ACM on Human-Computer Interaction, 3*(CSCW), Article 198, 1-29. https://doi.org/10.1145/3359300

Kalia, N. N. (1979). Images of men and women in Indian textbooks. *Comparative Education Review*, 23(2), 209-223.https://www.jstor.org/stable/1187923

King, T. L., Scovelle, A. J., Meehl, A., Milner, A. J., & Priest, N. (2021). Gender stereotypes and biases in early childhood: A systematic review. *Australasian Journal of Early Childhood, 46*(2), 112–125.

King, T. L., Scovelle, A. J., Meehl, A., Milner, A. J., & Priest, N. (2021). Gender stereotypes and biases in early childhood: A systematic review. *Australasian Journal of Early Childhood, 46*(2), 157–171. https://doi.org/10.1177/1836939121999849

Luckett, L., & Shay, S. (2020). *Reframing the Curriculum: A Transformative Approach*. *Critical Studies in Education, 61*(1), 50-65. https://doi.org/10.1080/17508487.2016.1227673

Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Humphrey, L., Basoalto, D., ... & Grundmann, M. (2019). MediaPipe: A Framework for Building Per

McHugh, M. L. (2013). The chi-square test of independence. *Biochemia Medica*, 23(2), 143-149. https://doi.org/10.11613/BM.2013.018

Meister, N., Zhao, D., Wang, A., Ramaswamy, V. V., Fong, R., & Russakovsky, O. (2022). Gender Artifacts in Visual Datasets. arXiv preprint arXiv:2206.09191.

Menon, S., & Vondrick, C. (2022). Visual Classification via Description from Large Language Models. *International Conference on Learning Representations (ICLR 2023)*. https://openreview.net/forum?id=jlCjycxcKc

Nandi, S., Bhattacharyya, S., & Das, S. (2020). Gender biased in textbook's picture of West Bengal board of primary education. *International Journal of Research*

National Council of Educational Research and Training. (2017). *Gender analysis of NCERT textbooks*. NCERT. https://ncert.nic.in/dgs/pdf/overallreportDGS_24_8_17.pdfAPPENDIX

Quinn, L., & Vorster, J. (2022). From Affirmative to Transformative Approaches: The Challenges of Academic Development. Studies in Higher Education, 47(3), 555-567. https://eric.ed.gov/?id=EJ1370564

Savarese, S., & Hoi, S. (2023). BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. *arXiv preprint arXiv:2301.12597*.

Signorella, M. L., Hayes, A. R., & Li, Y. (2020). Developmental differences in children's gender schemata about others: A meta-analytic review. *Developmental Review, 57*, 100910.

Taigman, Y., Yang, M., Ranzato, M. A., & Wolf, L. (2014). DeepFace: Closing the Gap to Human-Level Performance in Face Verification. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. Retrieved from https://doi.org/10.1109/CVPR.2014.220

Wolfe, R., & Caliskan, A. (2022). Markedness in Visual Semantic AI. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT '22)*, 1269-1279. https://doi.org/10.1145/3531146.3533179

Zhai, X., Kolesnikov, A., Houlsby, N., & Beyer, L. (2022). Scaling Vision Transformers. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 12094-12103. https://doi.org/10.1109/CVPR52688.2022.01179

Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., & Torralba, A. (2017). Places: A 10 million Image Database for Scene Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. Retrieved from https://doi.org/10.1109/TPAMI.2017.2723009

Zhou, K., Yang, J., Loy, C. C., & Liu, Z. (2022). Learning to prompt for vision-language models. *International Journal of Computer Vision*, 130(9), 2337-2348.

Zhu, X., Zhu, B., Tan, Y., Wang, S., Hao, Y., & Zhang, H. (2024). Enhancing zero-shot vision models by label-free prompt distribution learning and bias correcting. *arXiv preprint arXiv:2410.19294*. https://doi.org/10.48550/arXiv.2410.19294

 Bem, S. L. (1981). Gender schema theory: A cognitive account of sex typing. *Psychological Review, 88*(4), 354–364. https://doi.org/10.1037/0033-295X.88.4.354

 Bhog, D. (2002). Gender and curriculum. *Economic and Political Weekly, 37*(17), 1638-1642.

 Bigler, R. S., & Liben, L. S. (2007). Developmental intergroup theory: Explaining and reducing children's social stereotyping and prejudice. *Current Directions in Psychological Science, 16*(3), 162–166. https://doi.org/10.1111/j.1467-8721.2007.00496.x

 Ha, D., & Eck, D. (2018). A Neural Representation of Sketch Drawings. arXiv preprint arXiv:1704.03477.

 Jocher, G., Chaurasia, A., & Qiu, J. (2023). YOLO by Ultralytics. Retrieved from https://github.com/ultralytics/ultralytics

 Kokkola, L. (2010). [Review of *Power, voice and subjectivity in literature for young readers*, by M. Nikolajeva]. *International Research in Children's Literature, 3*(1), 106-109. https://doi.org/10.3366/ircl.2010.0016

 Lin, T.-Y., Maire, M., Belongie, S., Bourdev, L., Girshick, R., Hays, J., ... & Zitnick, C. L. (2014). Microsoft COCO: Common Objects in Context. arXiv preprint arXiv:1405.0312.

 Mayoux, L. (2001). Tackling the down side: Social capital, women's empowerment, and micro-finance in Cameroon. *Development and Change, 32*(3), 435-464. https://doi.org/10.1111/1467-7660.00213

 Nashriyah, N., Maulida, R., Ningsih, Y. S., & Yusuf, Y. Q. (2020). Addressing gender bias issues in elementary school EFL textbooks: An analysis of *Grow with English*. *Humanities & Social Sciences Reviews, 8*(3), 56-63. https://doi.org/10.18510/hssr.2020.837

 Nazli, S. (2003). [Review of *Paradigms of learning: The total literacy campaign in India*, by M. Karlekar]. *The Pakistan Development Review, 42*(2), 167-169. https://www.jstor.org/stable/41260534

 Sadker, D. M., & Zittleman, K. (2007). Practical strategies for detecting and correcting gender bias in your classroom. In D. Sadker & E. S. Silber (Eds.), *Gender in the classroom: Foundations, skills, methods, and strategies across the curriculum* (pp. 259–275). Lawrence Erlbaum Associates Publishers.

 Singh, M. (1998). Gender issues in children's literature. *ERIC Digest*.

 Smith, D. E. (1987). *The everyday world as problematic: A feminist sociology.* Northeastern University Presss

 Spitz, E. H. (1999). *Inside picture books*. Yale University Press.

 Syed, G. K., & Agha, N. (2019). Representation of gender in the secondary school Sindhi textbooks of Pakistan. *Asian Journal of Social Science, 47*(4-5), 566-580. https://www.jstor.org/stable/26848810

 Taylor, F. (2003). Content analysis and gender stereotypes in children's books. *Teaching Sociology, 31*(3), 300-311. https://doi.org/10.2307/3211327

 Trautner, H. M., Ruble, D. N., Cyphers, L., & Kirsten, B. (2005). Rigidity and flexibility of gender stereotypes in children: Developmental or differential? *Infant and Child Development, 14*(4), 365-381. https://doi.org/10.1002/icd.399

 Varghese, N., & Kumar, N. (2020). Femvertising as a media strategy to increase self-esteem of adolescents: An experiment in India. *Children and Youth Services Review, 113*, 104965. https://doi.org/10.1016/j.childyouth.2020.104965

 Wang, A., Liu, A., Zhang, R., Kleiman, A., Kim, L., Zhao, D., Shirai, I., Narayanan, A., & Russakovsky, O. (2021). REVISE: A tool for measuring and mitigating bias in visual datasets. *arXiv preprint arXiv:2004.07999*. https://doi.org/10.48550/arXiv.2004.07999